Linear Programming and the Simplex Method

Optimization Methods — seminar

Plan of the seminar

Part 1. Theory (45 min, no proofs)

1. What is a linear program (LP)
2. Standard form and geometry
3. The simplex method step by step
4. A full example by hand
5. Special cases, two-phase method, complexity

Part 2. Practice (45 min)

1. Graphical solution
2. `scipy.optimize.linprog`
3. Implement the tableau simplex method
4. Test it on random LPs
5. Bonus: Klee–Minty, two-phase simplex

Linear programming

Where linear programs appear

- **Production planning.** Choose how much of each product to make to maximize profit, given limited resources (materials, machine hours, labour).
- **Diet problem.** Choose amounts of foods to minimize cost while meeting nutrition requirements.
- **Transportation.** Ship goods from warehouses to stores at minimum cost, respecting supplies and demands.
- **Also:** scheduling, network flows, portfolio and blending problems, relaxations of integer problems.

Common structure: the objective is linear, every constraint is a linear equality or inequality, and variables are continuous.

Running example: a factory

A factory makes two products. Profit per unit: 3 and 5. Three resources are limited.

$$
\begin{aligned}
\max \quad & 3x_1 + 5x_2 & \\
\text{s.t.} \quad & x_1 \le 4 & \text{(resource 1)} \\
& 2x_2 \le 12 & \text{(resource 2)} \\
& 3x_1 + 2x_2 \le 18 & \text{(resource 3)} \\
& x_1,\ x_2 \ge 0 &
\end{aligned}
$$

- x_1, x_2 are the decision variables (amounts produced).
- We will solve this problem three ways: graphically, with the simplex method by hand, and with code.
- Answer (for reference): $x^* = (2, 6)$, profit 36.

General and standard form

General LP: minimize or maximize a linear function subject to linear $\leq$, $\geq$, $=$ constraints; variables may be free or sign-constrained.

Standard form (the one algorithms work with):

$$\min_{x \in \mathbb{R}^n} \; c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \qquad A \in \mathbb{R}^{m \times n}, \; \operatorname{rank} A = m.$$

Conversion rules

- $\max c^\top x \;\to\; \min(-c)^\top x$
- $a^\top x \leq \beta \;\to\; a^\top x + s = \beta, \; s \geq 0$ (slack variable)
- $a^\top x \geq \beta \;\to\; a^\top x - s = \beta, \; s \geq 0$ (surplus variable)
- free variable $x_j \;\to\; x_j = x_j^+ - x_j^-$, $x_j^\pm \geq 0$
- $b_i < 0 \;\to$ multiply row i by -1

The running example with slack variables

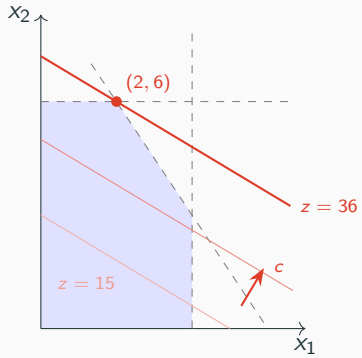
Add one slack per inequality: s_i = unused amount of resource i.

$$\begin{aligned} \max \quad & 3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 + s_1 = 4 \\ & 2x_2 + s_2 = 12 \\ & 3x_1 + 2x_2 + s_3 = 18 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{aligned}$$

- Now $m = 3$ equations and $n = 5$ variables.
- The matrix of the system is $[A \mid I]$: the slack columns form an identity matrix. This will give us a starting point for free.

Geometry: graphical solution

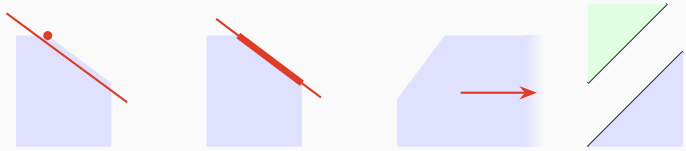
- Each inequality is a half-plane; the feasible set is their intersection, a **convex polygon** (in general, a polyhedron).
- Level lines $3x_1 + 5x_2 = z$ are parallel. Increasing z moves the line in the direction $c = (3, 5)$.
- Push the line as far as possible while it still touches the feasible set: the last contact point is $(2, 6)$, $z^* = 36$.



Key fact: optimum at a vertex

Fundamental theorem of LP (without proof)
If an LP in standard form has an optimal solution, then it has an optimal solution at a **vertex** of the feasible polyhedron.

Four possible outcomes:



Vertices = basic feasible solutions

- Pick m variables whose columns in A are linearly independent: the **basis** B. The other $n - m$ variables are **nonbasic**.
- Set all nonbasic variables to 0 and solve $A_B x_B = b$ for the basic ones.
- If $x_B \geq 0$, the point is a **basic feasible solution (BFS)**. Fact: BFS $\Leftrightarrow$ vertex.

Example. Basis $\{s_1, s_2, s_3\}$: $x_1 = x_2 = 0$, $s = (4, 12, 18) \Rightarrow$ vertex $(0, 0)$.
Basis $\{s_1, x_2, x_1\}$: $s_2 = s_3 = 0 \Rightarrow x_2 = 6$, $x_1 = 2$, $s_1 = 2 \Rightarrow$ vertex $(2, 6)$.

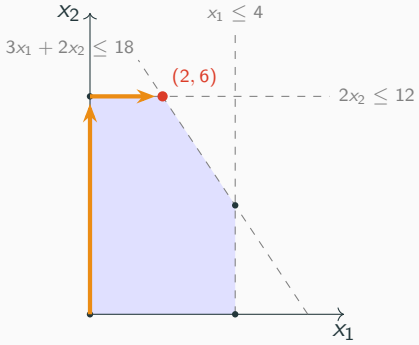
Problem: there can be up to $\binom{n}{m}$ bases (for $n = 100$, $m = 50$: about 10^{29}). Enumerating all vertices is hopeless — we need a smarter search.

The simplex method

Idea of the simplex method

- Start at some vertex (a BFS).
- Look at the edges leaving the vertex. If moving along some edge increases the objective, walk along it to the neighbouring vertex.
- Repeat. If no edge improves the objective, stop: by convexity, the current vertex is **globally optimal**.

In algebra: *moving to a neighbouring vertex = swapping one variable in the basis for one outside it*.



Setting: the problem we solve

$$\max\ c^\top x \quad \text{s.t.} \quad Ax \le b, \quad x \ge 0, \qquad \text{with } b \ge 0.$$

- Add slacks: $Ax + s = b$, $s \ge 0$.
- Because $b \ge 0$, the point $x = 0,\ s = b$ is feasible. It is a BFS with basis = all slacks.
- So we have a starting vertex for free. (The case $b \not\ge 0$ is handled by the two-phase method later.)

The algorithm in four steps, repeated until stop:

0. **Build** the initial tableau (once).
1. **Optimality check**: can the objective still grow?
2. **Choose the entering variable**: which edge to walk along.
3. **Ratio test**: how far we can walk; which variable leaves.
4. **Pivot**: update the tableau to the new vertex.

Step 0. The simplex tableau

Write the constraints and the objective $z - c^\top x = 0$ in one table:

$$T = \left[\begin{array}{cc|c} A & I & b \\ \hline -c^\top & 0 & 0 \end{array}\right] \qquad \text{size } (m+1) \times (n+m+1).$$

For the running example:

basis	x_1	x_2	s_1	s_2	s_3	b
s_1	1	0	1	0	0	4
s_2	0	2	0	1	0	12
s_3	3	2	0	0	1	18
z	-3	-5	0	0	0	0

- Every row is an equation: e.g. the last row reads $z - 3x_1 - 5x_2 = 0$.
- We also store which variable is basic in each row.

How to read a tableau

The tableau is always kept in a special (canonical) form:

- the column of each basic variable is a unit vector (1 in its row, 0 elsewhere, including the z row);
- the right-hand side column satisfies $b \geq 0$.

Then reading off the current vertex is trivial:

1. Nonbasic variables $= 0$.
2. Basic variable of row i = right-hand side b_i of that row.
3. The bottom-right entry = current objective value z.
4. Bottom row entries r_j are the **reduced costs**. The z row reads

$$z + \sum_{j \text{ nonbasic}} r_j x_j = \bar{z} \quad \Longrightarrow \quad z = \bar{z} - \sum_{j \text{ nonbasic}} r_j x_j.$$

Meaning of r_j: if we increase the nonbasic x_j by one unit, z changes by $-r_j$.

Step 1. Optimality check

From the z row: $z = \bar{z} - \sum_{j \text{ nonbasic}} r_j x_j$.

- All nonbasic variables are currently 0 and may only increase ($x \geq 0$).
- If $r_j \geq 0$ for every j: increasing any nonbasic variable cannot increase z.
- Hence no feasible point is better than the current one.

Stopping rule
If all entries of the z row (except the right-hand side) are ≥ 0, the current vertex is **optimal**. Stop and read the answer.

Otherwise there is some $r_j < 0$: increasing x_j increases z at rate $|r_j|$ per unit. Go to Step 2.

In the initial example tableau $r = (-3, -5, 0, 0, 0)$: both x_1 and x_2 would improve the profit.

Step 2. Choose the entering variable

Choose a column j with $r_j < 0$. The variable x_j will **enter** the basis: we start to increase it from 0. Geometrically: we pick an edge out of the current vertex.

Pivoting rules

- **Dantzig's rule:** the most negative r_j — the steepest improvement per unit. Usually few iterations.
- **Bland's rule:** the smallest index j with $r_j < 0$. Slower in practice but **never cycles**.
- Others: largest total improvement, steepest edge (used in real solvers).

Example. $r = (-3, -5, 0, 0, 0)$. Dantzig: x_2 enters (-5 is the most negative). Bland: x_1 enters.

Step 3. Ratio test: how far can we go?

Increase the entering variable: $x_j = t \geq 0$, other nonbasic variables stay 0. Row i reads $x_{B(i)} + a_{ij}\, t = b_i$, so

$$x_{B(i)}(t) = b_i - a_{ij}\, t \;\geq\; 0 \quad \text{must hold for every row } i.$$

- If $a_{ij} \leq 0$: $x_{B(i)}$ does not decrease — no limit from this row.
- If $a_{ij} > 0$: $x_{B(i)}$ hits zero at $t = b_i / a_{ij}$.

Minimum ratio rule

$$t^* = \min_{i:\ a_{ij} > 0} \frac{b_i}{a_{ij}}$$

The row i^* attaining the minimum is the **pivot row**; its basic variable becomes 0 and **leaves** the basis.

- Taking a larger t would make $x_{B(i^*)}$ negative — infeasible.
- If no $a_{ij} > 0$: t can grow forever and $z \to +\infty$. The LP is **unbounded**. Stop.
- Ties: pick the row with the smallest basic-variable index (Bland).

Step 4. Pivot

Goal: make x_j basic in row i^*, i.e. turn column j into a unit vector with 1 in row i^*. These are Gaussian elimination row operations:

1. **Normalize the pivot row:** $R_{i^*} \leftarrow R_{i^*} / a_{i^* j}$.
2. **Eliminate column j from every other row** k (the z row too):

$$R_k \leftarrow R_k - a_{kj}\, R_{i^*}.$$

3. **Update the basis:** $B(i^*) \leftarrow j$.

Why it is correct:

- Row operations do not change the set of solutions of the system.
- The new right-hand side is exactly $x_B(t^*)$, so it stays ≥ 0 (ratio test).
- The new z row expresses z through the new nonbasic variables; its corner value grows by $|r_j|\, t^*$.

Then go back to Step 1.

The algorithm on one slide

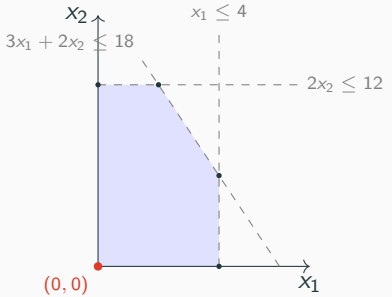
0. Build $T = [A\ I\ b;\ -c^\top\ 0\ 0]$, basis = slacks.
1. If all $r_j \geq 0 \Rightarrow$ **optimal**: basic variables $= b$, the rest $= 0$, $z^* =$ corner entry.
2. Pick entering column j with $r_j < 0$ (Dantzig or Bland).
3. Over rows with $a_{ij} > 0$ take $i^* = \arg\min b_i/a_{ij}$. If there is no such row $\Rightarrow$ **unbounded**.
4. Pivot on a_{i^*j}: normalize row i^*, eliminate column j elsewhere, set $B(i^*) = j$. Go to 1.

- Every iteration moves to a neighbouring vertex with z not smaller.
- If $t^* > 0$ at every step, z strictly increases, no vertex repeats, and the method stops after finitely many steps.

Worked example

Iteration 1: choose the pivot

	x_1	x_2	s_1	s_2	s_3	b	ratio
s_1	1	0	1	0	0	4	—
s_2	0	2	0	1	0	12	$12/2 = \mathbf{6}$
s_3	3	2	0	0	1	18	$18/2 = 9$
z	-3	-5	0	0	0	0	



1. Optimality: $r_1 = -3 < 0$, $r_2 = -5 < 0 \Rightarrow$ not optimal.
2. Entering (Dantzig): x_2, the most negative reduced cost.
3. Ratio test on column x_2: row s_1 has $a = 0$ (no limit); minimum ratio 6 in row $s_2 \Rightarrow s_2$ leaves. Pivot element $= 2$.

Interpretation: produce product 2 until resource 2 runs out ($x_2 = 6$); resource 3 would allow $x_2 = 9$.

Iteration 1: pivot

Row operations (pivot on row s_2, column x_2):

$$
\begin{aligned}
R_2 &\leftarrow R_2/2 && = (0,\ 1,\ 0,\ \tfrac{1}{2},\ 0 \mid 6) \\
R_1 &\leftarrow R_1 - 0 \cdot R_2 && = (1,\ 0,\ 1,\ 0,\ 0 \mid 4) \\
R_3 &\leftarrow R_3 - 2 \cdot R_2 && = (3,\ 0,\ 0,\ -1,\ 1 \mid 6) \\
R_z &\leftarrow R_z + 5 \cdot R_2 && = (-3,\ 0,\ 0,\ \tfrac{5}{2},\ 0 \mid 30)
\end{aligned}
$$

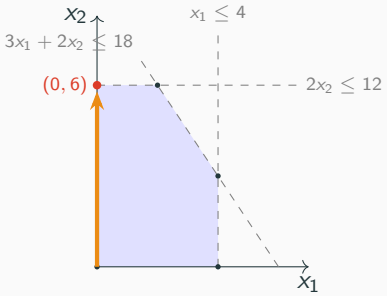
	x_1	x_2	s_1	s_2	s_3	b
s_1	1	0	1	0	0	4
x_2	0	1	0	$\frac{1}{2}$	0	6
s_3	3	0	0	-1	1	6
z	-3	0	0	$\frac{5}{2}$	0	30

New vertex: $x_1 = 0$, $x_2 = 6$
($s_1 = 4$, $s_3 = 6$), $z = 30$.

Check: z grew by $|r_2| \cdot t^* = 5 \cdot 6 = 30$.

Iteration 2: choose the pivot

	x_1	x_2	s_1	s_2	s_3	b	ratio
s_1	1	0	1	0	0	4	$4/1 = 4$
x_2	0	1	0	$\frac{1}{2}$	0	6	—
s_3	3	0	0	-1	1	6	$6/3 = \mathbf{2}$
z	-3	0	0	$\frac{5}{2}$	0	30	



1. Optimality: $r_1 = -3 < 0 \Rightarrow$ not optimal yet.
2. Entering: x_1 (the only negative reduced cost).
3. Ratio test: $\min(4/1,\ 6/3) = 2$ in row $s_3 \Rightarrow s_3$ leaves. Pivot element $= 3$.

Interpretation: produce product 1 until resource 3 is used up; resource 1 would allow 4.

Iteration 2: pivot

Row operations (pivot on row s_3, column x_1):

$$
\begin{aligned}
R_3 &\leftarrow R_3/3 && = (1,\ 0,\ 0,\ -\tfrac{1}{3},\ \tfrac{1}{3} \mid 2)\\
R_1 &\leftarrow R_1 - 1\cdot R_3 && = (0,\ 0,\ 1,\ \tfrac{1}{3},\ -\tfrac{1}{3} \mid 2)\\
R_2 &\leftarrow R_2 - 0\cdot R_3 && = (0,\ 1,\ 0,\ \tfrac{1}{2},\ 0 \mid 6)\\
R_z &\leftarrow R_z + 3\cdot R_3 && = (0,\ 0,\ 0,\ \tfrac{3}{2},\ 1 \mid 36)
\end{aligned}
$$

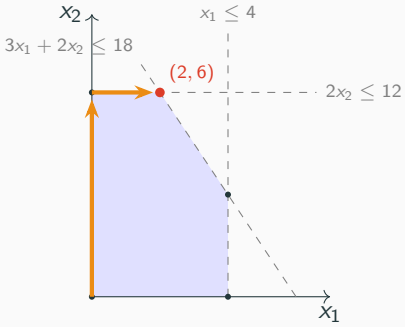
	x_1	x_2	s_1	s_2	s_3	b
s_1	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
x_2	0	1	0	$\frac{1}{2}$	0	6
x_1	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
z	0	0	0	$\frac{3}{2}$	1	36

New vertex: $x_1 = 2$, $x_2 = 6$ ($s_1 = 2$), $z = 36$.

Check: z grew by $3 \cdot 2 = 6$.

Optimality and the answer

- z row: $(0, 0, 0, \frac{3}{2}, 1)$ — all $\geq 0 \Rightarrow$ **optimal**.
- Read the answer: $x_1^* = 2$, $x_2^* = 6$, $z^* = 36$.
- $s_1 = 2$: two units of resource 1 are left unused; resources 2 and 3 are fully used.
- Path: $(0, 0) \to (0, 6) \to (2, 6)$ — two iterations, along two edges.



Bonus observation: shadow prices
The z row reads $z = 36 - \frac{3}{2}s_2 - s_3$. One more unit of resource 2 (allowing $s_2 = -1$) adds $\frac{3}{2}$ to the profit, one more unit of resource 3 adds 1, resource 1 adds 0. These are the **shadow prices** (dual variables).

Special cases and practical remarks

Unbounded problem in the tableau

max $x_1 + x_2$ s.t. $x_1 - x_2 \le 1,\ x \ge 0$.

	x_1	x_2	s_1	b
s_1	1	-1	1	1
z	-1	-1	0	0

x_1 enters, s_1 leaves $\longrightarrow$

	x_1	x_2	s_1	b
x_1	1	-1	1	1
z	0	-2	1	1

- Now x_2 should enter ($r_2 = -2$), but its column has **no positive entries**.
- Increasing $x_2 = t$: $x_1 = 1 + t \ge 0$ for every t, and $z = 1 + 2t \to \infty$.
- The ray $x(t) = (1 + t,\ t)$ is a feasible direction of unbounded improvement.

Multiple optima and infeasibility

Multiple optimal solutions

- At the optimal tableau some *nonbasic* variable has $r_j = 0$.
- Pivoting it in changes the vertex but not z: the whole edge between the two vertices is optimal.
- Example: $\max 3x_1 + 2x_2$ with the same constraints — the objective is parallel to $3x_1 + 2x_2 \leq 18$, every point between $(2, 6)$ and $(4, 3)$ is optimal.

Infeasible problem

- The basic method never detects it: it starts from $x = 0$, which requires $b \geq 0$.
- If b has negative entries or there are equality / $\geq$ constraints, $x = 0$ may be infeasible.
- Solution: the **two-phase method** (next slide).

Two-phase method

Phase I: find a feasible vertex

1. Write constraints as $Ax = b$ with $b \geq 0$ (flip rows with negative right-hand side).
2. Add an artificial variable $a_i \geq 0$ to every row: $Ax + a = b$. Then $x = 0,\ a = b$ is a BFS.
3. Solve $\min \sum_i a_i$ with the simplex method (first zero out the reduced costs of the basic a_i).
4. If the optimum is $> 0 \Rightarrow$ the original LP is **infeasible**.

Between phases

- Pivot remaining artificial variables out of the basis (they are at 0); drop redundant rows; delete artificial columns.

Phase II: optimize

- Put the original objective into the z row, make it canonical (zero reduced costs of basic variables), run the simplex method.

Degeneracy and cycling

- **Degenerate vertex:** some basic variable equals 0 (more constraints than necessary pass through the vertex).
- Then the ratio test may give $t^* = 0$: the basis changes but the point and z do not move.
- With an unlucky rule the method can return to an old basis and **cycle forever**.

Bland's rule (without proof)
Choose the entering variable with the smallest index among $r_j < 0$; break ties in the ratio test by the smallest index of the leaving variable. Then the simplex method always terminates.

In practice: real solvers use perturbation or other anti-cycling techniques; cycling on real problems is rare, but numerical tolerances matter.

Complexity

- **Worst case is exponential.** Klee–Minty cube: a deformed n-dimensional cube where Dantzig's rule visits all 2^n vertices, i.e. $2^n - 1$ iterations.
- No known pivoting rule is proven to be polynomial.
- **In practice** the simplex method is very fast: typically a small multiple of m iterations.
- **Polynomial algorithms** exist: the ellipsoid method (Khachiyan, 1979) and interior-point methods (Karmarkar, 1984).
- Modern solvers (HiGHS in `scipy`, Gurobi, CPLEX) implement both the (dual) simplex method and interior-point methods.

You will reproduce the Klee–Minty effect in the bonus task.

Practice

Practice: the notebook

File: `lp_simplex_seminar_student.ipynb`. Fill in every `# YOUR CODE HERE`; each task ends with a test cell.

Task	What to do	Time
0	Graphical solution: objective at vertices, level line	5 min
1	`linprog`: the board example and a furniture factory	8 min
2	`make_tableau`: build $[A\ I\ b;\ -c^\top\ 0\ 0]$	5 min
3	`choose_entering`, `ratio_test`, `pivot`	12 min
4	`run_simplex`, `simplex`; plot the path over vertices	8 min
5	Test on 200 random LPs; compare Dantzig and Bland	2 min
6*	Klee–Minty: iterations vs. dimension	homework
7*	Two-phase simplex for arbitrary b	homework

Implementation hints

- Store the tableau as a `numpy` array `T` of shape $(m+1, n+m+1)$ and the basis as a list `basis` with `basis[i]` = index of the basic variable of row `i`.
- Compare with zero using a tolerance: `EPS = 1e-9`.
- Ratio test: use only rows with `T[i, j] > EPS`; return `None` if there are none.
- Pivot: normalize the row first, then update *all* other rows including the last one.
- `linprog` **minimizes**: pass `-c` and report `-res.fun`.
- Current point: `x = zeros(n+m); x[basis] = T[:-1, -1]`, then take the first n entries.

Questions to think about

1. Why is it enough to look only at the vertices of the feasible set?
2. What does a column with no positive entries in the ratio test mean geometrically?
3. Why does starting from $x = 0$ work only when $b \geq 0$?
4. Dantzig's rule needs fewer iterations on random problems. Why do we still need Bland's rule?
5. In the final tableau, what do the numbers under the slack columns mean for the factory?